

Econometrics III

Bachelor in Econometrics and Data Science

Bachelor in Econometrics and Operations Research

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Vector Autoregressive Models

This Course: Econometrics 3

1. Multiple equations in regression models (week 1)
2. Panel data models (weeks 2 and 3)
 - ▷ Introduction and Random Effects model
 - ▷ Fixed Effects model and some extensions
3. Vector autoregressive models (weeks 3 and 4)
 - ▷ Stable VAR(1), dynamic properties, forecasting
 - ▷ Granger causality, impulse response, estimation, Stable VAR(p)
4. Non-stationary, cointegration, vector error correction (week 5)
5. Principal components, dynamic factor models (week 6)

Vector Autoregressive Models

- ▶ VAR(1) model
- ▶ Wold representation
- ▶ Dynamic properties
- ▶ Forecasting
- ▶ Granger causality
- ▶ Impulse response function
- ▶ Parameter estimation
 - ▷ Conditional regression estimation
 - ▷ Maximum likelihood estimation
 - ▷ Hypothesis testing
- ▶ Generalization to VAR(p) model
- ▶ What p to chose ?

Autoregressive model

Let's y_t represent a time series process for discrete time index t .

The time series process is defined for all $t \in \mathbb{Z}$,
that is $t = \dots, -2, -1, 0, 1, 2, \dots$

A realization of the time series y_t , of length T , is then for $t \in \mathbb{N}$,
that is $t = 1, \dots, T$.

An example for a univariate time series is the autoregressive process

$$y_t = \omega + \phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z},$$

where ω is the intercept, ϕ is the autoregressive coefficient and ε_t is an IID noise sequence.

Basics of autoregressive model

The basics for the autoregressive process

$$y_t = \omega + \phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z},$$

cannot be summarized in a few slides ... so we refer to the “time series course” in period 1 or any other reference that covers

univariate time series models.

Motivation for multivariate direction

- ⇒ Analyzing multiple time series simultaneously:
provides powerful tools to explore *dynamic inter-relations*
between economic variables.
- ⇒ It was mainly his research on VAR models that provided
Christopher Sims with the Nobel Prize in Economics in 2011.
- ⇒ VAR model has become the workhorse in macroeconometrics.

Literature

- ▶ Key readings:
 - ▷ H. Luetkepohl, New Introduction to Multiple Time Series Analysis
 - ▷ B. H. Baltagi, Econometric Analysis of Panel Data
- ▶ Also recommended:
 - ▷ J. D. Hamilton, Time Series Analysis
 - ▷ J. M. Wooldridge, Econometric Analysis of Cross Section and Panel Data
 - ▷ H. Pesaran, Time Series and Panel Data Econometrics

Multiple time series

Our goal: generalize univariate concepts, and study dynamic interactions amongst a set of time series variables collected in an $N \times 1$ vector

$$y_t = \begin{pmatrix} y_{1t} \\ y_{2t} \\ \vdots \\ y_{N,t} \end{pmatrix}.$$

⇒ Why is this interesting?

Why multiple time series analysis?

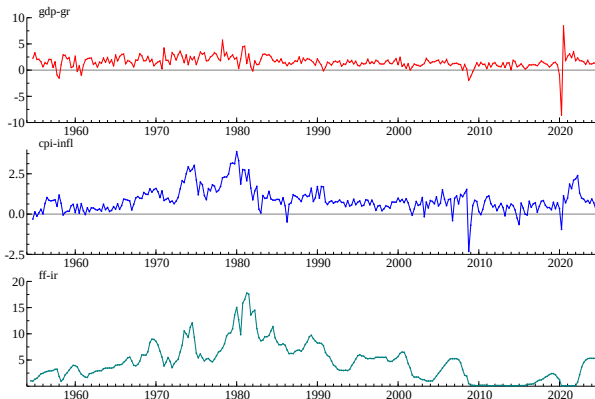
1. Structural analysis

- ▷ Explore dynamic interrelations between multiple variables.
- ▷ Typical questions: What is the dynamic effect of a change in one variable in the system (e.g. change in income) on other variables (e.g. consumption, investment)?
- ▷ Are relations implied by a multiple time series model consistent with economic theory?

2. Forecasting

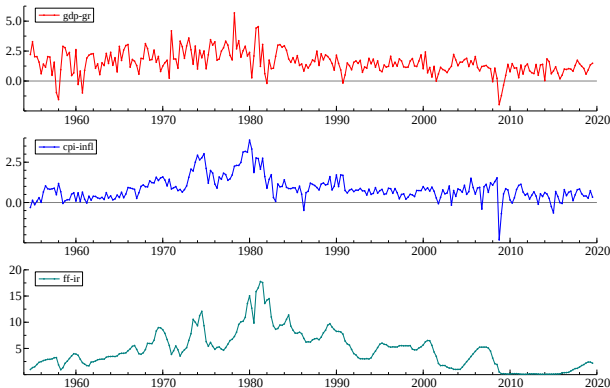
- ▷ Past data from several variables may contain information on future developments of a variable of interest, e.g. household consumption expenditures today may depend on past expenditures, but also on past income, interest rates, etc.
- ▷ The additional information content for household consumption expenditures is made an endogenous part of the model.

An illustration



(i) U.S. data on gross domestic product (GDP) quarterly growth, (ii) headline CPI inflation, (iii) Federal funds rate: quarterly time series 1954Q3 to 2024Q3.

An illustration: pre-COVID



(i) U.S. data on gross domestic product (GDP) quarterly growth, (ii) headline CPI inflation, (iii) Federal funds rate: quarterly time series 1954Q3 to 2019Q1.

Matrices, eigenvalues and eigenvectors

The eigenvalues of an $n \times n$ matrix A are defined as the roots of the characteristic equation

$$|A - \lambda I| = 0,$$

where $|A|$ is the determinant of matrix A . There are n distinctive eigenvalues for a non-singular matrix A .

For each eigenvalue λ , there exists a “unique” vector x such that

$$Ax = \lambda x,$$

where $n \times 1$ vector x is referred to as the eigenvector of A , associated with the eigenvalue λ .

Eigenvector x is a direction, it can be scaled to any magnitude.

An illustration

We have $n = 3$ and matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

The eigenvalues of matrix A are

$$\lambda_1 = 3.618, \quad \lambda_2 = 1.382, \quad \lambda_3 = 1,$$

and the eigenvectors are

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 0.618 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} -0.618 \\ -0.618 \\ 1 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

Eigenvalue analysis

Since we have

$$A x = \lambda x,$$

it follows in our $n = 3$ illustration that

$$A x_1 = \lambda_1 x_1, \quad A x_2 = \lambda_2 x_2, \quad A x_3 = \lambda_3 x_3,$$

and

$$A [x_1, x_2, x_3] = [x_1, x_2, x_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \Leftrightarrow A X = X \Lambda,$$

with obvious notation.

In case all eigenvalues are strictly non-zero, $\lambda_i \neq 0$, for $i = 1, \dots, n$, matrix A is not rank-deficient and its inverse exists.

Eigen-decomposition of a matrix

Hence, the eigen analysis of A leads us to the identity

$$AX = X\Lambda,$$

from which it follows that original matrix A can be decomposed as

$$A = X\Lambda X^{-1}$$

provided that A is not rank-deficient. The reverse result is

$$\Lambda = X^{-1}AX,$$

which implies that the eigenvectors allow the transformation of a full matrix A into a diagonal matrix Λ .

Other matrix decompositions can also be used for this purpose, including those of “Schur” and “Jordan”.

Complex eigenvalues

Since the eigenvalue is a solution of a characteristic function of higher orders, the solution can typically be in the complex range.

Even for $n = 2$, $|A - \lambda I| = 0$ is a polynomial of degree $n = 2$, which is often referred to as the characteristic polynomial of A . Hence, even this quadratic function can have complex roots.

We do not discuss this issue further as we need to move on. However, when VAR models are used in economic data modeling, the existence of complex-valued eigenvalues is the rule rather than an exception. In fact, they suggest “business cycle” dynamics.

VAR(1)

Let's start with the VAR(1) model which is the multivariate version of the AR(1) model

$$y_t = \omega + \phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z},$$

where ϕ is the autoregressive coefficient and ε_t is an IID noise sequence with mean zero and variance σ^2 .

From now on,

- ▶ y_t is an $N \times 1$ vector,
- ▶ ω is an $N \times 1$ vector,
- ▶ ϕ is replaced by the autoregressive coefficient $N \times N$ matrix Φ ,
- ▶ ε_t is an $N \times 1$ noise vector,
- ▶ σ^2 is replaced by the $N \times N$ variance matrix Σ .

VAR(1)

Consider two time series in vector y_t , $N = 2$, the vector autoregressive time series process of order 1 is given by

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z},$$

which is very similar to the univariate case, but now we have vectors and matrices,

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

where most notation is implied and clear, and where $\varepsilon_t \sim \text{NID}(0, \Sigma)$ with

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}.$$

Stable VAR(1) model

To investigate the statistical dynamic properties of the VAR(1) model,

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z},$$

we want to be assured that the dynamic process y_t is stationary.

For this purpose, we diagonalize the matrix Φ using an *eigenvalue decomposition*:

$$\Phi = C \Lambda C^{-1} \quad \Leftrightarrow \quad C^{-1} \Phi C = \Lambda,$$

where Λ is a *diagonal* $N \times N$ matrix with eigenvalues (usually ordered by size) of Φ while the columns of C consist of the corresponding eigenvectors.

Eigenvalue decomposition

The *eigenvalue decomposition* for a matrix can be powerful, see above.

Return to your *linear algebra* course to refresh your knowledge about eigenvalues and eigenvectors !!

A nice property (also from a computational viewpoint) is for powers of Φ :

$$\Phi = C \Lambda C^{-1}, \quad \Phi^2 = C \Lambda C^{-1} C \Lambda C^{-1} = C \Lambda^2 C^{-1},$$

where Λ^2 is simply the squares of its *diagonal* elements only.

More generally

$$\Phi^h = C \Lambda^h C^{-1}, \quad h = 1, 2, 3, \dots$$

From VAR(1) model to N AR(1) models

Given the transformation

$$C^{-1}\Phi C = \Lambda \quad \Rightarrow \quad C^{-1}\Phi = \Lambda C^{-1},$$

we next consider the following transformation for the observation vector y_t and its VAR(1) model, that is

$$z_t := C^{-1}y_t = C^{-1}\omega + C^{-1}\Phi y_{t-1} + C^{-1}\varepsilon_t, \quad t \in \mathbb{Z}.$$

It follows that z_t is generated by the model

$$z_t = \omega^* + \Lambda z_{t-1} + \varepsilon_t^*,$$

where $\omega^* = C^{-1}\omega$ and $\varepsilon_t^* = C^{-1}\varepsilon_t$ with $\varepsilon_t^* \sim \text{NID}(0, C^{-1}\Sigma C^{-1'})$.

The eigenvalues $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$ are important as they determine the dynamic properties of z_t , and partly $y_t = C z_t$.

Stable AR models $\Rightarrow$ Stable VAR model

The model for the $N \times 1$ time series vector z_t is also VAR(1)

$$z_t = \omega^* + \Lambda z_{t-1} + \varepsilon_t^*,$$

but since Λ is a diagonal matrix, each element of z_t can be described as an *univariate* AR(1) model, that is

$$z_{j,t} = \omega_j^* + \lambda_j z_{j,t-1} + \varepsilon_{j,t}^*, \quad j = 1, \dots, N,$$

with obvious notation.

When all individual time series processes in z_t are stationary, then all individual time series processes in y_t are also stationary, since $y_t = C z_t$ which implies that each process in y_t is a linear combination of stationary processes. Hence, y_t is also stationary, see Appendix with “General Theorem on Stationarity”.

Stable VAR(1) process

The conclusion is that the stationarity of y_t requires the stationarity of z_t which simply requires the “stability” conditions

$$-1 < \lambda_i < 1, \quad \forall i = 1, \dots, N,$$

where λ_i is the i th eigenvalue of matrix Φ in the VAR(1) model for y_t , that is

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z}.$$

It is a simple procedure:

- ▶ Compute the eigenvalues of Φ , that is solve the characteristic equation $|\Phi - \lambda I| = 0$, obtaining N distinctive roots λ_i .
- ▶ When all eigenvalues $-1 < \lambda_i < 1$, for $i = 1, \dots, N$, we can conclude that the VAR(1) model is stable.

Dynamic properties VAR(1) process

Once the stable VAR(1) process is established, we can explore the statistical (dynamic) properties of the VAR(1) process.

We can do this most effectively via the Wold decomposition of the VAR(1) process.

The Wold decomposition can be obtained by simple substitution:

$$\begin{aligned}y_1 &= \omega + \Phi y_0 + \varepsilon_1, \\y_2 &= \omega + \Phi y_1 + \varepsilon_2 \\&= \omega + \Phi (\omega + \Phi y_0 + \varepsilon_1) + \varepsilon_2 \\&= (I + \Phi) \omega + \Phi^2 y_0 + \Phi \varepsilon_1 + \varepsilon_2 \\y_3 &= (I + \Phi + \Phi^2) \omega + \Phi^3 y_0 + \Phi^2 \varepsilon_1 + \Phi \varepsilon_2 + \varepsilon_3, \quad \text{etc.}\end{aligned}$$

Dynamic properties VAR(1) process

By continuing the substitution, we obtain

$$y_t = (I + \Phi + \dots + \Phi^{t-1})\omega + \Phi^t y_0 + \sum_{j=0}^{t-1} \Phi^j \varepsilon_{t-j},$$

and get effectively the Wold decomposition as $t \rightarrow \infty$. We have

$$\Phi^t = C \Lambda^t C^{-1}, \quad t = 1, 2, 3, \dots,$$

and

$$I + \Phi + \dots + \Phi^{t-1} = C[I + \Lambda + \dots + \Lambda^{t-1}]C^{-1}.$$

Dynamic properties VAR(1) process

The Wold decomposition is

$$y_t = (I + \Phi + \dots + \Phi^{t-1})\omega + \Phi^t y_0 + \sum_{j=0}^{t-1} \Phi^j \varepsilon_{t-j}, \quad \text{with } t \rightarrow \infty.$$

A stable VAR(1) has $|\lambda_j| < 1$, for $j = 1, \dots, N$, and as $t \rightarrow \infty$

$$\Phi^t = C \Lambda^t C^{-1} \rightarrow 0, \quad \text{as } t \rightarrow \infty,$$

since $\lambda_j^t \rightarrow 0$, $\forall j$, as $t \rightarrow \infty$.

Dynamic properties VAR(1) process

The Wold decomposition is based on

$$y_t = (I + \Phi + \dots + \Phi^{t-1})\omega + \sum_{j=0}^{t-1} \Phi^j \varepsilon_{t-j}, \quad \text{with } t \rightarrow \infty,$$

A stable VAR(1) has $|\lambda_j| < 1$, for $j = 1, \dots, N$, and as $t \rightarrow \infty$,

$$I + \Phi + \dots + \Phi^{t-1} = C[I + \Lambda + \dots + \Lambda^{t-1}]C^{-1} \rightarrow C(I - \Lambda)^{-1}C^{-1} \quad \text{as } t \rightarrow \infty,$$

since $1 + \lambda_j + \lambda_j^2 + \dots = 1 / (1 - \lambda_j)$.

Further, we have $(AB)^{-1} = B^{-1}A^{-1}$ for full rank matrices A and B , then

$$C(I - \Lambda)^{-1}C^{-1} = [C(I - \Lambda)C^{-1}]^{-1} = (I - \Phi)^{-1},$$

with $A = C(I - \Lambda)$ and $B = C^{-1}$ since $A^{-1} = (I - \Lambda)^{-1}C^{-1}$. Hence,

$$(I + \Phi + \dots + \Phi^{t-1})\omega = (I - \Phi)^{-1}\omega.$$

Wold representation of VAR(1) process

These results imply that the Wold decomposition of the stable VAR(1) process

$$y_t = (I + \Phi + \Phi^2 + \dots)\omega + \sum_{j=0}^{\infty} \Phi^j \varepsilon_{t-j},$$

reduces to

$$y_t = \mu + \sum_{j=0}^{\infty} \Phi^j \varepsilon_{t-j},$$

where

$$\mu = (I - \Phi)^{-1}\omega.$$

Mean and variance of stable VAR(1)

The Wold representation of the stable VAR(1)

$$y_t = \mu + \sum_{j=0}^{\infty} \Phi^j \varepsilon_{t-j},$$

can be used to show that the mean of a VAR(1) is

$$\mathbf{E}(y_t) := \mu = (I - \Phi)^{-1} \omega,$$

since $\mathbf{E}(\varepsilon_t) = 0$, $\forall t$. The variance matrix for y_t is

$$\text{Var}(y_t) := \Gamma(0) = \sum_{j=0}^{\infty} \Phi^j \Sigma \Phi^{j'}.$$

Variance of stable VAR(1) process

$$\begin{aligned}
 \Gamma(0) &= \sum_{j=0}^{\infty} \Phi^j \Sigma \Phi^{j'} \\
 &= C \left[\sum_{j=0}^{\infty} \Lambda^j C^{-1} \Sigma C^{-1'} \Lambda^j \right] C' \\
 &= C \left[\sum_{j=0}^{\infty} \Lambda^j \Sigma^* \Lambda^j \right] C', \quad \text{with } \Sigma^* = C^{-1} \Sigma C^{-1'}.
 \end{aligned}$$

When we vec as in [K4], we have $\text{vec}(\Sigma^*) = (C^{-1} \otimes C^{-1})\text{vec}(\Sigma)$ and

$$\begin{aligned}
 \text{vec}[\Gamma(0)] &= (C \otimes C) \text{vec} \left(\sum_{j=0}^{\infty} \Lambda^j \Sigma^* \Lambda^j \right) = (C \otimes C) \sum_{j=0}^{\infty} \text{vec}(\Lambda^j \Sigma^* \Lambda^j) \\
 &= (C \otimes C) \sum_{j=0}^{\infty} (\Lambda^j \otimes \Lambda^j) \text{vec}(\Sigma^*) \\
 &= (C \otimes C) (I - \Lambda \otimes \Lambda)^{-1} (C^{-1} \otimes C^{-1}) \text{vec}(\Sigma).
 \end{aligned}$$

Variance of stable VAR(1) process

Finally, since $(AB)^{-1} = B^{-1}A^{-1}$ and [K3] $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$:

$$\begin{aligned}
 \text{vec}[\Gamma(0)] &= (C \otimes C)(I - \Lambda \otimes \Lambda)^{-1}(C^{-1} \otimes C^{-1})\text{vec}(\Sigma) \\
 &= [(I - \Lambda \otimes \Lambda)(C^{-1} \otimes C^{-1})]^{-1}(C^{-1} \otimes C^{-1})\text{vec}(\Sigma) \\
 &= [(C \otimes C)(I - \Lambda \otimes \Lambda)(C^{-1} \otimes C^{-1})]^{-1}\text{vec}(\Sigma) \\
 &= (I - C\Lambda C^{-1} \otimes C\Lambda C^{-1})^{-1}\text{vec}(\Sigma) \\
 &= (I - \Phi \otimes \Phi)^{-1}\text{vec}(\Sigma),
 \end{aligned}$$

Last steps are valid partly due to [K2] $(\bar{A} \otimes \bar{B})(\bar{C} \otimes \bar{D}) = (\bar{AC} \otimes \bar{BD})$.

For a stationary VAR(1) process with given coefficient matrices Φ and Σ , we have analytical expressions for mean μ and variance $\Gamma(0)$.

In a similar way we can derive analytic expressions for the autocovariance matrices $\Gamma(\tau) := \text{Cov}(y_t, y_{t-\tau})$ for $\tau = 1, 2, 3, \dots$

Mean and variance of stationary VAR(1)

We have derived the dynamic properties under the assumptions of a stable VAR(1), that is $|\lambda_j| < 1, \forall j$. The realizations of time series variables in y_t from a stable VAR(1) are stationary since moments are time-invariant (they do not depend on time t).

We can also derive the dynamic properties of a VAR(1) process under the assumption of stationarity. For example, in a stationary setting the mean of y_t is time-invariant (constant over time), we have

$$\mu := E(y_t) = \omega + \Phi E(y_{t-1}) + E(\varepsilon_t) = \omega + \Phi \mu,$$

and $\mu = (I - \Phi)^{-1}\omega$. The variance of a stationary VAR(1) process is

$$\Gamma(0) := \text{Var}(y_t) = \Phi \text{Var}(y_{t-1})\Phi' + \Sigma = \Phi\Gamma(0)\Phi' + \Sigma,$$

which we can solve directly as $\text{vec}[\Gamma(0)] = (I - \Phi \otimes \Phi)^{-1}\text{vec}(\Sigma)$.

Autocovariances of stationary VAR(1)

The autocovariances can also be derived in this way.

For example, $\Gamma(1) := \text{Cov}(y_t, y_{t-1}) = \mathbf{E}[(y_t - \mu)(y_{t-1} - \mu)']$

$$\begin{aligned}\Gamma(1) &= \Phi \mathbf{E}[(y_{t-1} - \mu)(y_{t-1} - \mu)'] + \mathbf{E}[\varepsilon_t (y_{t-1} - \mu)'] \\ &= \Phi \Gamma(0) + 0,\end{aligned}$$

we have $\Gamma(1) = \Phi \Gamma(0)$. More generally, it can be shown that

$$\Gamma(2) = \Phi \Gamma(1), \quad \Gamma(3) = \Phi \Gamma(2), \quad \dots$$

or consequently

$$\Gamma(\tau) = \Phi^\tau \Gamma(0), \quad \tau = 0, 1, 2, \dots$$

You see: deriving properties under the stationary assumption is easier !

Autocorrelations of stationary VAR(1)

The auto-crosscorrelations for elements of y_t from a VAR(1) process are typically defined as

$$\rho_{ij}(\tau) := \Gamma_{ij}(\tau) / \sqrt{\Gamma_{ii}(0) \cdot \Gamma_{jj}(0)},$$

where $\Gamma_{ij}(\tau)$ is the (i, j) element of matrix $\Gamma(\tau)$, for $i, j = 1, \dots, N$ and $\tau = 0, 1, 2, \dots$

When $i = j$, we have that $\rho_{ii}(\tau)$ is the more usual autocorrelation.

When $i \neq j$, we have that $\rho_{ij}(\tau)$ is the auto-crosscorrelation.

Dynamic properties VAR(1): Example 1

Consider VAR(1) model, with $N = 2$,

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.3 \end{pmatrix} + \begin{bmatrix} 0.8 & 0.1 \\ -0.2 & 0.5 \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

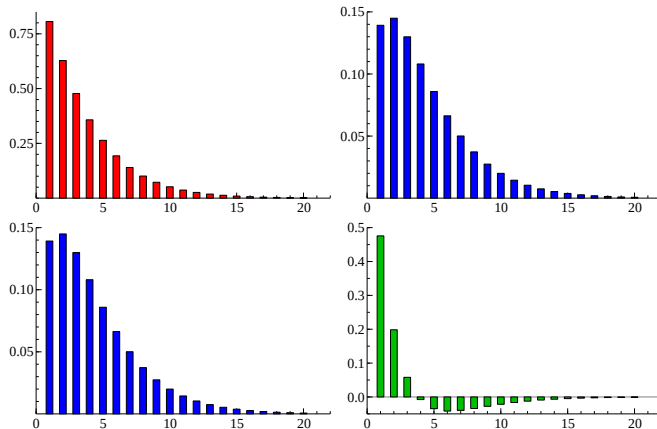
with

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

we then have

$$\mu = \begin{pmatrix} 2.75 \\ -0.5 \end{pmatrix}, \quad \Gamma(0) = \begin{bmatrix} 5.79 & 0.35 \\ 0.35 & 2.88 \end{bmatrix}.$$

Dynamic properties VAR(1): Example 1



Dynamic properties VAR(1): Example 2

Consider VAR(1) model, with $N = 2$,

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.3 \end{pmatrix} + \begin{bmatrix} 0.8 & 0.0 \\ -0.2 & 0.5 \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

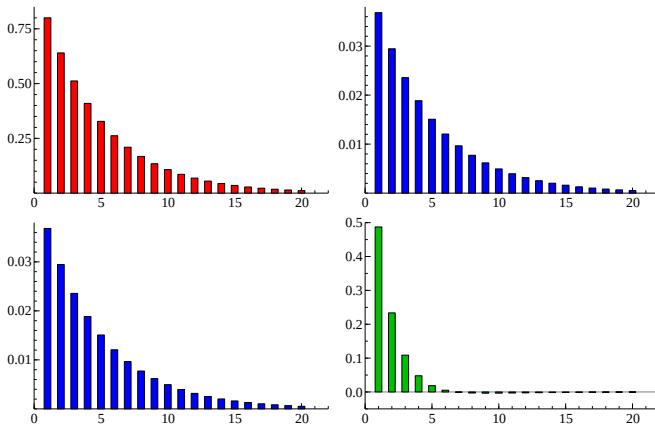
with

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

we then have

$$\mu = \begin{pmatrix} 3.0 \\ -0.6 \end{pmatrix}, \quad \Gamma(0) = \begin{bmatrix} 5.556 & 0.185 \\ 0.185 & 2.914 \end{bmatrix}.$$

Dynamic properties VAR(1): Example 2



Forecasting



Forecasting for a stable VAR(1)

Forecasting is about taking conditional expectations and variances. We define the one-step ahead point-forecast as

$$\hat{y}_{t+1|1:t} := E(y_{t+1}|y_1, \dots, y_t),$$

where the conditioning of $y_1, \dots, y_t$ implies that we consider all these y_j 's to be realized and fixed (NOT being subject to random variation).

For given VAR(1) coefficients ω , Φ and Σ , we derive point-forecast as

$$\begin{aligned}\hat{y}_{t+1|1:t} &= E(y_{t+1}|y_1, \dots, y_t) \\ &= E(\omega + \Phi y_t + \varepsilon_{t+1}|y_1, \dots, y_t) \\ &= \omega + \Phi E(y_t|y_1, \dots, y_t) + E(\varepsilon_{t+1}|y_1, \dots, y_t) \\ &= \omega + \Phi y_t.\end{aligned}$$

Notice $y_1, \dots, y_t$ rely on $\varepsilon_1, \dots, \varepsilon_t$ but not on ε_{t+1} , and all ε 's are IID, with zero mean. This forecast is very easy to compute.

Forecast uncertainty

The forecast $\hat{y}_{t+1|1:t}$ is subject to uncertainty or *stochastic variation* due to ε_{t+1} since the forecast error is

$$e_{t+1|1:t} := y_{t+1} - \hat{y}_{t+1|1:t} = y_{t+1} - \omega - \Phi y_t = \varepsilon_{t+1}.$$

The amount of uncertainty is typically measured by the corresponding conditional variance, that is

$$\begin{aligned} F_1 &:= \text{Var}(y_{t+1}|y_1, \dots, y_t) \\ &= E\{[y_{t+1} - E(y_{t+1}|y_1, \dots, y_t)][y_{t+1} - E(y_{t+1}|y_1, \dots, y_t)]' | y_1, \dots, y_t\} \\ &= E[(y_{t+1} - \hat{y}_{t+1|1:t})(y_{t+1} - \hat{y}_{t+1|1:t})' | y_1, \dots, y_t] \\ &= E(\varepsilon_{t+1}\varepsilon_{t+1}' | y_1, \dots, y_t) \\ &= E(\varepsilon_{t+1}\varepsilon_{t+1}') = \text{Var}(\varepsilon_{t+1}) \\ &= \Sigma. \end{aligned}$$

Forecast uncertainty

For a given forecast $\hat{y}_{t+1|1:t}$, we have forecast error ε_{t+1} and variance Σ .

A forecast confidence interval for $\hat{y}_{t+1|1:t}$ is based on Σ , typically the diagonal elements of Σ are used for this.

With a certain confidence level, the realization y_{t+1} will then be within the forecast confidence interval.

There are more sources of forecast uncertainty:

- ▶ Parameter uncertainty: choice of coefficients for ω , Φ and Σ when computing forecasts can be different than those used for the realizations of $y_1, \dots, y_t$ (data generation process, DGP) since they are not known, see Estimation part later.
- ▶ Model uncertainty: it is also uncertain whether the VAR(1) model has been used to generate the realizations of $y_1, \dots, y_t$ in the first place, see VAR(p) model part later.

Multi-step forecasting with a VAR(1)

Multi-step forecasting is often even of more interest to policy makers. We define the multi-step ahead point-forecast as

$$\hat{y}_{t+h|1:t} := E(y_{t+h}|y_1, \dots, y_t), \quad h = 1, 2, 3, \dots$$

Let's start with $h = 2$:

$$\begin{aligned}\hat{y}_{t+2|1:t} &= E(y_{t+2}|y_1, \dots, y_t) \\ &= E(\omega + \Phi y_{t+1} + \varepsilon_{t+2}|y_1, \dots, y_t) \\ &= \omega + \Phi E(y_{t+1}|y_1, \dots, y_t) + E(\varepsilon_{t+2}|y_1, \dots, y_t) \\ &= \omega + \Phi \hat{y}_{t+1|1:t} \\ &= \omega + \Phi(\omega + \Phi y_t) \\ &= (I + \Phi)\omega + \Phi^2 y_t.\end{aligned}$$

Then do $h = 3$ and repeat the above, do you obtain the result below ?

$$\hat{y}_{t+3|1:t} = (I + \Phi + \Phi^2)\omega + \Phi^3 y_t.$$

Multi-step forecasting with a VAR(1)

Generally, the multi-step forecasting function is given by

$$\hat{y}_{t+h|1:t} = (I + \Phi + \Phi^2 + \dots + \Phi^{h-1})\omega + \Phi^h y_t, \quad h = 1, 2, 3, \dots$$

This result follows from repeatedly substitution as we also have done before for the Wold decomposition.

We notice that as $h \rightarrow \infty$, we have

$$\hat{y}_{t+h|1:t} = (I + \Phi + \Phi^2 + \dots)\omega + \Phi^h y_t \rightarrow \mu, \text{ as } h \rightarrow \infty,$$

and, indeed, it's the unconditional mean $\mu = E(y_t) = (I - \Phi)^{-1}\omega$.

Multi-step forecast error

Given the multi-step forecast

$$\hat{y}_{t+2|1:t} = (I + \Phi)\omega + \Phi^2 y_t,$$

for $y_{t+2} = \omega + \Phi y_{t+1} + \varepsilon_{t+2} = (I + \Phi)\omega + \Phi^2 y_t + \Phi \varepsilon_{t+1} + \varepsilon_{t+2}$, it follows immediately that the corresponding forecast error is given by

$$e_{t+2|1:t} = \Phi \varepsilon_{t+1} + \varepsilon_{t+2},$$

and more generally

$$e_{t+h|1:t} = \sum_{j=0}^{h-1} \Phi^j \varepsilon_{t+h-j}, \quad h = 1, 2, 3, \dots$$

Multi-step forecast uncertainty

Given the multi-step forecast error

$$e_{t+h|1:t} = \sum_{j=0}^{h-1} \Phi^j \varepsilon_{t+h-j}, \quad h = 1, 2, 3, \dots,$$

the corresponding multi-step forecast uncertainty is given by

$$F_h = \sum_{j=0}^{h-1} \Phi^j \Sigma \Phi^{j'}, \quad h = 1, 2, 3, \dots$$

We notice that as $h \rightarrow \infty$, we have that F_h converges to the unconditional variance

$$\text{vec}[F_h] \rightarrow \text{vec}[\Gamma(0)] = (I - \Phi \otimes \Phi)^{-1} \text{vec}(\Sigma), \quad h \rightarrow \infty.$$

Granger causality

Granger causality is due to Granger (1969).

Consider two time series variables y_{1t} and y_{2t} , then y_{2t} is said to Granger-cause y_{1t} if y_{1t} can be predicted more efficiently if the information from the past and present of y_{2t} is taken into account, in addition to the past and present of y_{1t} itself.

This definition is rather intriguing and “food for thought”. However, in the context of the VAR(1) model with $N = 2$, that is

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

it reduces to a simple question: is ϕ_{12} equal to zero or not ?

Granger causality

Granger causality for a VAR(1) model reduces to a simple notion:

when $\phi_{12} = 0$, y_{2t} does not Granger-cause y_{1t} . Otherwise it does !

- ▶ When $\phi_{21} = 0$, y_{1t} does not Granger-cause y_{2t} .
- ▶ When matrix Φ is unknown and must be estimated (see below), we can formally carry out the hypothesis test of $H_0 : \phi_{12} = 0$, where this null is NOT Granger-cause (so rejection is nice for Granger).
- ▶ Groups of variables can Granger-cause another group of variables: then multiple coefficients in Φ need zeroes, to NOT Granger-cause.
- ▶ In this case, but also for VAR(p) models, multiple zero restrictions are in place: statistical testing involves joint F-test (see below).

Impulse response

Impulse response functions represent the mechanisms through which shocks spread over time within a prespecified dynamic model.

In the case of a stable VAR(1) model,

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z},$$

we consider its Wold representation

$$y_t = \mu + \sum_{j=0}^{\infty} C_j \varepsilon_{t-j}, \quad \mu = (I - \Phi)^{-1} \omega, \quad C_j = \Phi^j.$$

Matrix C_j can be interpreted as the impact on y_t of a shock in ε_{t-j} .

Impulse response

Impulse response functions can also be defined more formally.

For the VAR(1) Wold representation

$$y_t = \mu + \sum_{j=0}^{\infty} C_j \varepsilon_{t-j}, \quad \mu = (I - \Phi)^{-1} \omega, \quad C_j = \Phi^j.$$

matrix C_j can be interpreted as

$$\frac{\partial y_t}{\partial \varepsilon'_{t-j}} = C_j, \quad j = 1, 2, 3, \dots$$

Element (i, k) of C_j identifies the impact of a unit increase in the k th element of ε_{t-j} for the i th element of y_t , while holding all other elements of ε fixed at zero.

Impulse response

However, the impulse response function is usually regarded as a forward-looking entity.

We then have, given the VAR(1) Wold representation,

$$y_t = \mu + \sum_{j=0}^{\infty} C_j \varepsilon_{t-j}, \quad \mu = (I - \Phi)^{-1} \omega, \quad C_j = \Phi^j.$$

matrix C_j being interpreted as

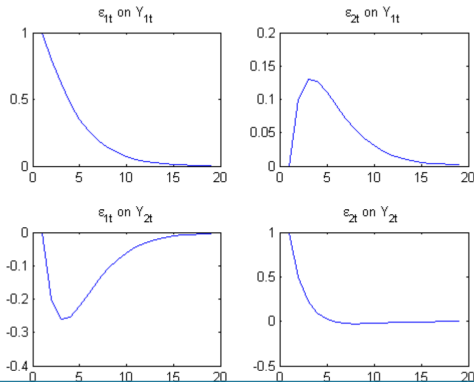
$$\frac{\partial y_{t+j}}{\partial \varepsilon'_t} = C_j, \quad j = 1, 2, 3, \dots$$

Element (i, k) of C_j identifies the impact of a unit increase in the k th element of ε_t for the i th element of y_{t+j} , while holding all other elements of ε fixed at zero.

Impulse response: Example 1

Consider VAR(1) model, with $N = 2$,

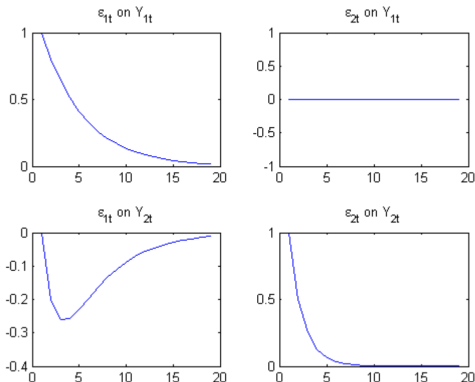
$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} 0.8 & 0.1 \\ -0.2 & 0.5 \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$



Impulse response: Example 2

Consider VAR(1) model where y_{2t} does not Granger-cause y_{1t} :

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} 0.8 & 0 \\ -0.2 & 0.5 \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$



Orthogonal impulse response

Impulse response functions represent the mechanisms through which shocks spread over time within a prespecified dynamic model.

However, the shocks in a VAR(1) model are dependent upon each other: a shock in one is instantaneously also a shock in another:

$$\varepsilon_t \sim \text{NID}(0, \Sigma),$$

where Σ is a “full” positive-definite matrix.

To create a “unique” shock, we can diagonalize Σ and consider the corresponding transformed noise term:

$$\Sigma = PP', \quad \eta_t = P^{-1}\varepsilon_t, \quad \text{such that } \text{Var}(\eta_t) = I.$$

A shock in an element of η_t is a “unique” shock.

Orthogonal impulse response

Given the orthogonality of η_t :

$$\Sigma = PP', \quad \eta_t = P^{-1}\varepsilon_t, \quad \text{such that } \text{Var}(\eta_t) = I,$$

the VAR(1) Wold representation can alternatively be written in terms of η 's:

$$y_t = \mu + \sum_{j=0}^{\infty} C_j \varepsilon_{t-j} = \mu + \sum_{j=0}^{\infty} C_j P P^{-1} \varepsilon_{t-j} = \mu + \sum_{j=0}^{\infty} C_j^* \eta_{t-j}.$$

Hence, the orthogonal impulse response is based on $C_j^* = C_j P$ rather than C_j itself.

Discussion

- ▶ Impulse response functions (IRF) provide a useful tool to understand the dynamic interrelated workings of a VAR(1).
- ▶ It is intensely used in empirical (monetary) macroeconomics and by policy makers at e.g. central banks.
- ▶ The concept is simple and is in essence a particular interpretation of the VAR(1) Wold representation coefficients.
- ▶ Orthogonal IRF is addressing the issue of the uniqueness of a shock in the VAR(1).
- ▶ However, the decomposition $\Sigma = PP'$ is not unique, there are infinitively many P 's that establish $PP' = \Sigma$.
- ▶ More later ...

Unknown parameters

So far, we have, more or less, implicitly assumed that the VAR(1) coefficients ω , Φ and Σ are known. In a practical setting, where we analyze real (or “realized”) time series, we have no clue about these coefficients, they are unknown to us. We even don’t know whether the VAR(1) model is the “true” or correct model for time series y_t .

Let us assume that the VAR(1) model is the correct specification for y_t , but that its coefficients are unknown to us. We need to estimate ω , Φ and Σ . We consider two methodologies for estimation:

- ▶ Least squares method
- ▶ Maximum likelihood method

Realized (or observed) time series

Parameter estimation can start once we have a set of realizations for y_t .

We do not change the notation for a realized y_t (this is perhaps a bit sloppy, sorry), we just assume that we have the following set of realizations (or observations) for y_t , that is

$$\{y_1, y_2, \dots, y_T\}, \quad \text{or} \quad \{y_t\}, \quad t \in \mathbb{N}.$$

When these realizations are generated by the VAR(1) model

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t,$$

we can only treat the observations $y_2, \dots, y_T$ as coming from the VAR(1) model since they depend on their corresponding previous observations.

As we do not have y_0 , we typically assume that observation y_1 is a realization of its *unconditional distribution*

$$y_1 \sim N[\mu, \Gamma(0)].$$

Conditional estimation

Hence, we assume that we have a set of observations:

$$\{y_1, y_2, \dots, y_T\},$$

We assume that they are generated by the VAR(1) model

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t = 2, \dots, T,$$

and we treat observation y_1 as coming from

$$y_1 \sim N[\mu, \Gamma(0)].$$

When we discard the “initial” distribution for y_1 in the estimation process, we refer to this as conditional estimation, and this is what we do below, unless we indicate otherwise.

Least squares method

The least squares method is associated with the regression model, so let's try to express the VAR(1) model

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t = 2, \dots, T,$$

as a set of regression equations. The i th equation of the VAR(1) model is given by

$$y_{it} = \omega_i + \bar{\phi}_i y_{t-1} + \varepsilon_{it}, \quad i = 1, \dots, N,$$

where $\bar{\phi}_i$ is the i th row of matrix Φ . The inner-product can be expressed as $\bar{\phi}_i y_{t-1} \equiv y'_{t-1} \bar{\phi}'_i$ such that

$$y_{it} = \omega_i + y'_{t-1} \bar{\phi}'_i + \varepsilon_{it}, \quad i = 1, \dots, N.$$

This does not yet look as a regression model !!

Least squares method

Given the VAR(1) model equation-by-equation,

$$y_{it} = \omega_i + y'_{t-1} \bar{\phi}_i + \varepsilon_{it}, \quad i = 1, \dots, N,$$

we can collect all observations over time, $t = 2, \dots, T$, in the $T - 1 \times 1$ observation vector $y_i^* = (y_{i2}, \dots, y_{iT})'$ and the $T - 1 \times N$ matrix $X = (y_1, \dots, y_{T-1})'$, to obtain the model

$$y_i^* = 1_{T-1} \cdot \omega_i + X \bar{\phi}_i + \varepsilon_i^*,$$

with $T - 1 \times 1$ vector $1_{T-1} = (1, \dots, 1)'$.

By defining $T - 1 \times N + 1$ matrix $\bar{X} = (1_{T-1}, X)$ and $N + 1 \times 1$ coefficient vector $\beta_i = (\omega_i, \bar{\phi}_i)'$, we obtain the multiple regression model

$$y_i^* = \bar{X} \beta_i + \varepsilon_i^*, \quad i = 1, \dots, N.$$

This looks familiar: SUR with same X , for each equation i !!

Least squares method

We have shown that we can express the VAR(1) model

$$y_t = \omega + \Phi y_{t-1} + \varepsilon_t, \quad t = 2, \dots, T,$$

as a SUR with the same X :

$$y_i^* = \bar{X} \beta_i + \varepsilon_i^*, \quad i = 1, \dots, N.$$

This opens the door for straightforward estimation of β_i : its GLS estimates are obtained by simple least squares, equation-by-equation.

Notice: estimation of the β_i 's is effectively the estimation of ω and Φ .

This is a powerful result and has made VAR modeling rather popular !!

Least squares method

The estimation procedure for a VAR(1) model is therefore simple:

- ▶ Consider the VAR(1) model, equation-by-equation, $i = 1, \dots, N$,

$$y_{it} = \omega_i + \bar{\phi}_i y_{t-1} + \varepsilon_{it}, \quad i = 1, \dots, N,$$

and apply the ordinary least squares method to each equation i (with summation over t) for the joint estimation of ω_i and $\bar{\phi}_i$.

- ▶ The residuals are simply constructed as

$$e_{it} = y_{it} - \hat{\omega}_i + y'_{t-1} \hat{\bar{\phi}}_i, \quad t = 1, \dots, T,$$

for each equation $i = 1, \dots, N$.

- ▶ The estimate of Σ is then obtained by

$$\hat{\Sigma} = \frac{1}{T - (N + 2)} \sum_{t=2}^T e_t e_t', \quad e_t = (e_{1t}, \dots, e_{Nt})'.$$

Maximum Likelihood estimation

Our treatment of estimation has so far been implied, arguing that we can regard the VAR(1) model as a specific SUR, with the same X . This implies that least squares estimation is “justified”.

A more formal approach is to consider a likelihood-based analysis for the VAR(1) model.

We adopt the typical approach of formulating the likelihood function in time series modeling: the prediction error decomposition.

The likelihood function is based on the joint density function $p(y_1, \dots, y_T)$ for the selection of stochastic variables $\{y_1, \dots, y_T\}$. Once we have a set of observations for $\{y_1, \dots, y_T\}$, we treat $p(y_1, \dots, y_T)$ as the *likelihood function* which is then only a function of the parameter vector ψ . We denote the likelihood function as $\ell(y_1, \dots, y_T; \psi)$.

Maximum Likelihood estimation

Examples of the prediction error decomposition are

$$T = 2 : p(y_1, y_2) = p(y_1) p(y_2|y_1),$$

$$\begin{aligned} T = 3 : p(y_1, y_2, y_3) &= p(y_1, y_2) p(y_3|y_1, y_2) \\ &= p(y_1) p(y_2|y_1) p(y_3|y_1, y_2), \end{aligned}$$

and in general we have

$$p(y_1, \dots, y_T) = p(y_1) \prod_{t=2}^T p(y_t|y_1, \dots, y_{t-1}).$$

Maximum Likelihood estimation

The prediction error decomposition $p(y_1) \prod_{t=2}^T p(y_t|y_1, \dots, y_{t-1})$ in logs is

$$\log p(y_1, \dots, y_T) = \log p(y_1) + \sum_{t=2}^T \log p(y_t|y_1, \dots, y_{t-1}).$$

In case of the VAR(1) model with a multivariate normally distributed IID sequence $\{\varepsilon_1, \dots, \varepsilon_T\}$, the predictive log-density at time t , $\log p(y_t|y_1, \dots, y_{t-1})$, is given by

$$-\frac{N}{2} \log 2\pi - \frac{1}{2} \log |\Sigma| - \frac{1}{2} (y_t - \Phi y_{t-1})' \Sigma^{-1} (y_t - \Phi y_{t-1}),$$

for $t = 2, \dots, T$. The initial log-density $\log p(y_1)$ is given by

$$-\frac{N}{2} \log 2\pi - \frac{1}{2} \log |\Gamma(0)| - \frac{1}{2} (y_1 - \mu)' \Gamma(0)^{-1} (y_1 - \mu).$$

Maximum Likelihood estimation

We typically ignore the initial log-density $\log p(y_1)$ because of convenience (see earlier) but also because this part introduces nonlinear functions of Φ due to μ and $\Gamma(0)$. This ignorance leads to what we refer to as the conditional likelihood.

Also, this approach leads to analytical estimation results which we are familiar with in regression. We can underline this by the following development

$$\omega + \Phi y_{t-1} \equiv BX_t,$$

where $B = [\omega, \Phi]$ and $X_t = (1, y'_{t-1})'$.

The VAR(1) conditional log-likelihood function $\ell_c(\psi)$ is then given by

$$-\frac{N(T-1)}{2} \log 2\pi - \frac{T-1}{2} \log |\Sigma| - \sum_{t=2}^T \frac{1}{2} (y_t - BX_t)' \Sigma^{-1} (y_t - BX_t).$$

Maximum Likelihood estimation

The least squares estimate for B in the VAR model is given by

$$\hat{B}_{LS} = \left(\sum_{t=2}^T y_t X_t' \right) \left(\sum_{t=2}^T X_t X_t' \right)^{-1}.$$

We show that $\hat{B}_{LS}$ is also the maximum likelihood estimator (MLE) of B . Define $e_t = y_t - \hat{B}_{LS} X_t$. The part of $\ell_c(\psi)$ that only involves B is

$$\begin{aligned} -2\ell_c &= \cdots + \sum_{t=2}^T (y_t - BX_t)' \Sigma^{-1} (y_t - BX_t) \\ &= \cdots + \sum_{t=2}^T [(y_t - \hat{B}_{LS} X_t) - (B - \hat{B}_{LS}) X_t]' \Sigma^{-1} [\dots] \\ &= \cdots + \sum_{t=2}^T [e_t - (B - \hat{B}_{LS}) X_t]' \Sigma^{-1} [e_t - (B - \hat{B}_{LS}) X_t]. \end{aligned}$$

Maximum Likelihood estimation

The B relevant part of the log-likelihood is

$$\sum_{t=2}^T [e_t - (B - \hat{B}_{LS})X_t]' \Sigma^{-1} [e_t - (B - \hat{B}_{LS})X_t],$$

and by multiplying it out, we have three scalars: $s1 = \sum_{t=2}^T e_t' \Sigma^{-1} e_t$ does not involve B ,

$$s2 = -2 \sum_{t=2}^T e_t' \Sigma^{-1} (B - \hat{B}_{LS})X_t,$$

turns out to be zero (see next page), and only relevant part for B is

$$s3 = \sum_{t=2}^T X_t' (B - \hat{B}_{LS})' \Sigma^{-1} (B - \hat{B}_{LS})X_t.$$

Since Σ (and Σ^{-1}) is positive definite, the smallest value for $s3$ is obtained clearly when $B = \hat{B}_{LS}$. Hence the MLE of B equals $\hat{B}_{LS}$.

Derivation that $s_2 = 0$

To show that scalar $s_2 = -2 \sum_{t=2}^T e'_t \Sigma^{-1} (B - \hat{B}_{LS}) X_t$ is zero, we have

$$\begin{aligned} s_2 = \text{trace}(s_2) &= -2 \text{trace} \left(\sum_{t=2}^T e'_t \Sigma^{-1} (B - \hat{B}_{LS}) X_t \right) \\ &= -2 \text{trace} \left(\left[\Sigma^{-1} (B - \hat{B}_{LS}) \right] \sum_{t=2}^T X_t e'_t \right) \end{aligned}$$

since $\text{trace}(AB) = \text{trace}(BA)$. Further,

$$\sum_{t=2}^T X_t e'_t = \sum_{t=2}^T X_t (y_t - \hat{B}_{LS} X_t)' = \sum_{t=2}^T X_t y'_t - \left(\sum_{t=2}^T X_t X'_t \right) \hat{B}'_{LS},$$

and after substitution of $\hat{B}_{LS} = \left(\sum_{t=2}^T y_t X'_t \right) \left(\sum_{t=2}^T X_t X'_t \right)^{-1}$, we have

$$s_2 = 0. \text{ Notice } \hat{B}'_{LS} = \left(\sum_{t=2}^T X_t X'_t \right)^{-1} \left(\sum_{t=2}^T X_t y'_t \right).$$

Maximum Likelihood estimation

Given that we have obtained $\hat{B}_{MLE} \equiv \hat{B}_{LS}$, we can define the *profile* log-likelihood function $\ell_c^p(\psi^*)$ as

$$-\frac{N(T-1)}{2} \log 2\pi - \frac{T-1}{2} \log |\Sigma| - \frac{1}{2} \sum_{t=2}^T e_t' \Sigma^{-1} e_t,$$

from which it follows from maximum likelihood regression theory that

$$\hat{\Sigma}_{MLE} = \frac{1}{T-1} \sum_{t=2}^T e_t e_t'.$$

The maximized log-likelihood value is therefore

$$-\frac{N(T-1)}{2} \log 2\pi - \frac{T-1}{2} \log |\hat{\Sigma}_{MLE}| - \frac{N(T-1)}{2}.$$

Maximum Likelihood estimation

The maximized log-likelihood value for the VAR model, given by

$$\hat{\ell}^{\max} = -\frac{N(T-1)}{2}(1 + \log 2\pi) - \frac{T-1}{2} \log |\hat{\Sigma}_{\text{MLE}}|,$$

or simply

$$\hat{\ell}^{\max} = \text{constant} - \frac{T-1}{2} \log |\hat{\Sigma}_{\text{MLE}}|,$$

is of use for hypothesis testing, model selection and information criteria, see below.

Maximum Likelihood estimation

MLE asymptotic theory applies to the VAR model straightforwardly.

For example, MLEs are consistent even if the true noise sequence of $\{\varepsilon_1, \dots, \varepsilon_T\}$ is non-Gaussian.

The MLE asymptotic properties for $\hat{B}_{\text{MLE}}$, without proofs, are:

- Consistency: $\text{plim}_{T \rightarrow \infty} \hat{B}_{\text{MLE}} = B$.
- Asymptotic Normality: $\sqrt{T} \text{vec}(\hat{B}_{\text{MLE}} - B) \xrightarrow{d} N(0, Z^{-1} \otimes \Sigma)$.

where $Z = \text{plim}_{T \rightarrow \infty} T^{-1} \sum_{t=2}^T X_t X_t'$.

Maximum Likelihood estimation

The Asymptotic Normality (AN) result

$$\sqrt{T} \text{vec}(\hat{B}_{\text{MLE}} - B) \xrightarrow{d} N(0, Z^{-1} \otimes \Sigma),$$

is used, in particular, for the approximate distribution of $\hat{B}_{\text{MLE}}$:

$$\text{vec}(\hat{B}_{\text{MLE}}) \stackrel{a}{\sim} N\left[\text{vec}(B), \bar{Z}_T^{-1} \otimes \hat{\Sigma}_{\text{MLE}}\right],$$

where $\bar{Z}_T = \sum_{t=2}^T X_t X_t'$.

Based on the AN result, we can compute approximate standard errors for $\hat{B}_{\text{MLE}}$, but we can also carry out hypothesis testing.

Hypothesis testing

A linear hypothesis with respect to B can be established by the null of

$$H_0 : R \text{vec}(B) = r,$$

where R is a known (selection) matrix with m rows (restrictions) and r is a known $m \times 1$ vector (often of zeroes).

The *likelihood-ratio* (LR) test is then constructed as

$$2(\hat{\ell}_1^{\max} - \hat{\ell}_0^{\max}) \stackrel{a}{\sim} \chi_m^2,$$

where $\hat{\ell}_1^{\max}$ is for the unrestricted and $\hat{\ell}_0^{\max}$ is for the restricted model.

When the “distance” between the two maximized profile log-likelihoods is large enough (relative to the χ_m^2 distribution and a given confidence level), we reject the null hypothesis.

Hypothesis testing

Apart from the *likelihood-ratio* (LR) test, we can also base hypothesis testing on

- ▶ Lagrange-Multiplier test, and
- ▶ Wald test.

We do not discuss the details, but the slides above contain all the information to construct these test statistics.

All likelihood-based tests can be adopted for linear hypotheses such as

$$H_0 : R \text{vec}(B) = r,$$

including those in case of Granger-causality testing.

Illustration: testing for Granger-causality

The linear restriction for the null-hypothesis

$$H_0 : R \text{vec}(B) = r,$$

can be used for Granger-causality testing

Consider the VAR(1) model

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}.$$

To verify whether y_{2t} does not Granger-cause y_{1t} , we need to test the hypothesis $H_0 : \phi_{12} = 0$. For this VAR(1), we have $B = \Phi$ and choose

$$R = (0, 0, 1, 0), \quad \text{vec}(B) = (\phi_{11}, \phi_{21}, \phi_{12}, \phi_{22})', \quad r = 0.$$

to construct the LR test for this null of y_{2t} does not Granger-cause y_{1t} .

Illustration: testing for Granger-causality

All likelihood-based tests can be adopted for linear hypotheses such as $H_0 : R \text{vec}(B) = r$, including those in case of Granger-causality testing.

However, in case VARs are treated as SURs, and estimated via regression, equation-by-equation, Granger-causality is usually tested using the standard F -test given by

$$F = \frac{T - k}{q} \cdot \frac{\text{RSS}_0 - \text{RSS}_1}{\text{RSS}_1},$$

where

- ▶ RSS_0 : residual sum of squares for the restricted model (under H_0)
- ▶ RSS_1 : residual sum of squares for the unrestricted model (under H_1)
- ▶ q : number of regressors in restricted model
- ▶ k : this number in unrestricted model (for VAR(1), $k = N + 1$)

Replacing unknown parameters by estimates

For a given set of time series observations $\{y_t\}$, $t \in \mathbb{N}$, estimation of the unknown parameters ω , Φ and Σ in a VAR(1) model can take place via the methods of regression or maximum likelihood.

In practice, we then replace the unknown parameters with their estimates. However, the estimates are subject to uncertainty and we need to allow for this. For example, we can provide *approximate standard errors* when parameter estimates are presented.

But also, forecast and impulse response functions are nonlinear functions of ω , Φ and Σ . Once we replace these by their estimates, forecasting and impulse responses are also subject to parameter uncertainty, but how to account for this in a nonlinear setting ?

Bootstrap method

Consider the VAR(1) model and collect all unknown coefficients in ω , Φ and Σ in the parameter vector ψ . Focus is on estimating a function $\vartheta(\psi)$ which can represent an impulse response function, point forecast, forecast standard error, autocovariance function, etc.

1. Estimate ψ to obtain $\hat{\psi}$ and compute function $\vartheta(\hat{\psi})$.
2. Select a set of bootstrap residuals $\varepsilon_1^*, \dots, \varepsilon_T^*$ randomly with replacement from demeaned residuals $\hat{\varepsilon}_t - \bar{\varepsilon}$, for $t = 2, \dots, T$.
3. Construct bootstrap time series using the selected bootstrap residuals:

$$y_t^* = \hat{\omega} + \hat{\Phi}y_{t-1}^* + \varepsilon_t^*, \quad t = 2, \dots, T$$

with $y_1^* \sim N[\hat{\mu}, \hat{\Gamma}(0)]$.

4. Estimate ψ to obtain $\hat{\psi}^*$ from the bootstrap time series, and compute function $\vartheta(\hat{\psi}^*)$.
5. Repeat steps 1. to 4. M times (for M large).

Bootstrap method

M bootstrap series of $\vartheta(\hat{\psi}^*)$ constitute M “realizations” of $\vartheta(\psi)$. From these realizations various sample statistics can be considered, including mean, variance, 10% quantile, 90% quantile, etc. In this way we can construct e.g 80% bootstrap confidence intervals.

This is a general and powerful method to allow for parameter uncertainty in, for example, impulse response functions.

The bootstrap method is computationally demanding and belongs to a wider class of Monte Carlo simulation methods.

In the day-to-day practice, the use of the bootstrap method is common for impulse responses, but less so for forecasting.

VAR(2)

Consider two time series in vector y_t , $N = 2$, the vector autoregressive time series process of order 2 is given by

$$y_t = \omega + \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \varepsilon_t, \quad t \in \mathbb{Z},$$

which we can write more explicitly as

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{1,11} & \phi_{1,12} \\ \phi_{1,21} & \phi_{1,22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \\ \begin{bmatrix} \phi_{2,11} & \phi_{2,12} \\ \phi_{2,21} & \phi_{2,22} \end{bmatrix} \begin{pmatrix} y_{1,t-2} \\ y_{2,t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

where most notation is implied and clear, and where $\varepsilon_t \sim \text{NID}(0, \Sigma)$.

Vector Autoregressive Model of order p

The vector autoregressive time series process of order p is given by

$$y_t = \omega + \Phi_1 y_{t-1} + \dots + \Phi_p y_{t-p} + \varepsilon_t, \quad t \in \mathbb{Z},$$

which is very similar to the univariate AR(p) model, but now we have vectors and matrices !

We refer to this model as the VAR(p) model.

We have considered the dynamic properties, forecasting, impulse responses, Granger-causality, estimation, etc. for a VAR(1) model, now we need to reconsider these topics for the VAR(p) model.

Are you ready for take-off ?

Companion form

The vector autoregressive time series process of order p as given by

$$y_t = \omega + \Phi_1 y_{t-1} + \dots + \Phi_p y_{t-p} + \varepsilon_t, \quad t \in \mathbb{Z},$$

can be written in companion form. In other words, we can represent the VAR(p) model as an expanded VAR(1) model:

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \end{pmatrix} = \begin{pmatrix} \omega \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I & 0 & \cdots & 0 & 0 \\ 0 & I & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I & 0 \end{bmatrix} \begin{pmatrix} y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \\ y_{t-p} \end{pmatrix} + \begin{pmatrix} \varepsilon_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Companion form

The vector autoregressive time series process of order p as given by

$$y_t = \omega + \Phi_1 y_{t-1} + \dots + \Phi_p y_{t-p} + \varepsilon_t, \quad t \in \mathbb{Z},$$

can be written in companion form as the VAR(1) model

$$Y_t = \varphi + \Psi Y_{t-1} + \xi_t,$$

where $\varphi = (\omega', 0, 0, \dots, 0)'$,

$$Y_t = \begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \end{pmatrix}, \quad \Psi = \begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I & 0 & \cdots & 0 & 0 \\ 0 & I & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I & 0 \end{bmatrix}, \quad \xi_t = \begin{pmatrix} \varepsilon_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Companion form

This is a very convenient development, we can write any VAR(p) model as a VAR(1).

Given that we have derived so many results for the VAR(1) model, can we adopt these results now also for the VAR(p) when expressed in companion form, as a VAR(1) ?

The answer is that we can indeed use the VAR(1) results to obtain expressions for the mean, variance, and autocovariance functions, and for the impulse response and forecasting functions, of the VAR(p) model.

So we are done ? Well, not completely, let's revisit a few topics that deserve some further comments.

Stable VAR(p)

The parameter conditions for a stable VAR(p) require all eigenvalues of the matrix Ψ of its companion form to have values between -1 and 1 :

$$\Psi = C \Lambda C^{-1} \quad \Leftrightarrow \quad C^{-1} \Psi C = \Lambda,$$

where Λ is a *diagonal* $Np \times Np$ matrix with eigenvalues (usually ordered by size) of Ψ while the columns of C consist of the corresponding eigenvectors. A stable VAR(p) process requires

$$|\lambda_i| < 1, \quad \forall i = 1, \dots, Np,$$

The eigenvalues are clearly functions of the matrices $\Phi_1, \dots, \Phi_p$, and given the specific structure of Ψ , the stability conditions can also be expressed in terms of the VAR(p) coefficients directly.

Given the potentially high-dimensional matrix Ψ , it may not be the most computationally efficient way to obtain the eigenvalues, but this is a matter for *computer science* !!

Computational issue: an illustration

Matrix Ψ is

$$\Psi = \begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I & 0 & \cdots & 0 & 0 \\ 0 & I & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I & 0 \end{bmatrix},$$

and its inverse is given by

$$\Psi^{-1} = \begin{bmatrix} 0 & I & 0 & \cdots & 0 \\ 0 & 0 & I & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & I \\ \Phi_1^- & \Phi_2^- & \cdots & \Phi_{p-1}^- & \Phi_p^- \end{bmatrix},$$

where $\Phi_1^- = \Phi_p^{-1}$ and $\Phi_i^- = -\Phi_p^{-1} \Phi_{i-1}$, for $i = 2, 3, \dots, p$. Verify this !

Computational issues

This illustration of Ψ^{-1} shows that although we adopt the companion form to conceptually derive results, the actual computations may appear high-dimensional, but there are many short-cuts to carry out the computations very efficiently, including computing Ψ^{-1} and eigenvalues λ_i of Ψ .

Mean and variance expressions

For a stable VAR(p), we consider its companion form, and obtain the mean expression:

$$\mu = E(y_t) = J E(Y_t) = J(I - \Psi)^{-1}\varphi = J(I - \Psi)^{-1}J'\omega.$$

since $y_t = J Y_t$ as we define $N \times Np$ selection matrix $J = (I_N, 0, \dots, 0)$. The variance expression from the companion form is

$$\text{vec}[\text{Var}(Y_t)] = (I - \Psi \otimes \Psi)^{-1}\text{vec}[\text{Var}(\xi_t)].$$

Since $\Gamma(0) = J \text{Var}(Y_t) J'$ and $\text{Var}(\xi_t) = J' \Sigma J$, we have.

$$\text{vec}[\text{Var}(\xi_t)] = (J \otimes J)'\text{vec}(\Sigma),$$

The variance expression is

$$\begin{aligned}\text{vec}[\Gamma(0)] &= \text{vec}[\text{Var}(y_t)] = (J \otimes J)\text{vec}[\text{Var}(Y_t)] = \\ &= (J \otimes J)(I - \Psi \otimes \Psi)^{-1}(J \otimes J)'\text{vec}(\Sigma).\end{aligned}$$

Autocovariance expressions

Actually, $\text{Var}(Y_t)$ contains the autocovariance matrices $\Gamma(i)$ for $i = 1, \dots, p-1$, that is for $p = 5$,

$$\text{Var}(Y_t) = \text{Var} \begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ y_{t-4} \end{pmatrix} = \begin{bmatrix} \Gamma(0) & \Gamma(1)' & \Gamma(2)' & \Gamma(3)' & \Gamma(4)' \\ \Gamma(1) & \Gamma(0) & \Gamma(1)' & \Gamma(2)' & \Gamma(3)' \\ \Gamma(2) & \Gamma(1) & \Gamma(0) & \Gamma(1)' & \Gamma(2)' \\ \Gamma(3) & \Gamma(2) & \Gamma(1) & \Gamma(0) & \Gamma(1)' \\ \Gamma(4) & \Gamma(3) & \Gamma(2) & \Gamma(1) & \Gamma(0) \end{bmatrix},$$

The remaining $\Gamma(i)$ for $i = p, p+1, \dots$ can be obtained via the so-called Yule-Walker equations, which we (quickly, in big steps) derive as follows

$$\begin{aligned} \Gamma(i) &= E(y_t^* y_{t-i}^{*'}) \\ &= E[(\Phi_1 y_{t-1}^{*'} + \dots + \Phi_p y_{t-p}^{*'}) y_{t-i}^{*'}] \\ &= \Phi_1 \Gamma(i-1) + \Phi_2 \Gamma(i-2) + \dots + \Phi_p \Gamma(i-p), \end{aligned}$$

for $i = p, p+1, \dots$, where $y_t^* = y_t - \mu$, $\forall t$.

Forecasting with a VAR(p)

For the VAR(p) model in companion form, the one-step ahead point-forecast is given by

$$\hat{Y}_{t+1|1:t} = \varphi + \Psi Y_t,$$

and

$$\hat{y}_{t+1|1:t} = J \hat{Y}_{t+1|1:t} = J(\varphi + \Psi Y_t) = \omega + B Y_t,$$

since $y_t = J Y_t$, $J\varphi = \omega$ and $J\Psi = B = [\Phi_1, \dots, \Phi_p]$.

This forecast is easy to compute: $\hat{y}_{t+1|1:t} = \omega + \Phi_1 y_t + \dots + \Phi_p y_{t-p+1}$.

The forecast error is

$$e_{t+1|1:t} := y_{t+1} - \hat{y}_{t+1|1:t} = y_{t+1} - \omega - B Y_t = \varepsilon_{t+1},$$

with its corresponding conditional (error) variance given by

$$F_1 = \text{Var}(y_{t+1}|y_1, \dots, y_t) = \text{Var}(\varepsilon_{t+1}) = \Sigma.$$

Multi-step forecasting with a VAR(p)

For VAR(p) in companion form, the multi-step forecasting function:

$$\hat{y}_{t+h|1:t} = J \hat{Y}_{t+h|1:t} = J(I + \Psi + \Psi^2 + \dots + \Psi^{h-1})\varphi + J\Psi^h Y_t,$$

for $h = 1, 2, 3, \dots$. The multi-step forecast error is

$$e_{t+h|1:t} = J \sum_{j=0}^{h-1} \Psi^j \xi_{t+h-j},$$

and the corresponding multi-step forecast uncertainty is given by

$$F_h = J \left[\sum_{j=0}^{h-1} \Psi^j \text{Var}(\xi_t) \Psi^{j'} \right] J',$$

for $h = 1, 2, 3, \dots$. These analytical expressions are correct and in terms of the companion form: their efficient computation needs to take account of the sparse nature of the matrices involved.

Granger causality

For example, in the VAR(2) model

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{1,11} & \phi_{1,12} \\ \phi_{1,21} & \phi_{1,22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \\ \begin{bmatrix} \phi_{2,11} & \phi_{2,12} \\ \phi_{2,21} & \phi_{2,22} \end{bmatrix} \begin{pmatrix} y_{1,t-2} \\ y_{2,t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

the hypothesis of “ y_{2t} does not Granger-cause y_{1t} ” now requires the null to be formulated as

$$H_0 : \phi_{1,12} = \phi_{2,12} = 0.$$

Also for the VAR(p) model, this hypothesis involves multiple coefficients ($H_0 : \phi_{1,12} = \phi_{2,12} = \dots = \phi_{p,12} = 0$).

We can adopt a likelihood-based test statistic, see above.

Impulse Response function

For a VAR(p) model, in companion form, the impulse response function for Y_t can be represented by C_j^Y in the Wold representation,

$$Y_t = E(Y_t) + \sum_{j=0}^{\infty} C_j^Y \xi_{t-j}, \quad E(Y_t) = (I - \Psi)^{-1} \varphi, \quad C_j^Y = \Psi^j.$$

The impulse response function for y_t is then represented by C_j and is based on

$$y_t = J Y_t = J(I - \Psi)^{-1} J' \omega + \sum_{j=0}^{\infty} J \Psi^j J' \varepsilon_{t-j}.$$

Hence,

$$C_j = J \Psi^j J', \quad j = 0, 1, 2, \dots$$

Maximum likelihood estimation

Our treatment of MLE was based on the model representation

$$y_t = BX_t + \varepsilon_t,$$

with $B = \Phi$ and $X_t = y_{t-1}$ for the VAR(1) model.

In case of the VAR(p) model, we can base MLE on the same model representation but now with

$$B = J\Psi = [\Phi_1, \dots, \Phi_p], \quad X_t = Y_{t-1},$$

which is directly obvious from its companion form.

The MLE treatment in the slides above then applies directly to the VAR(p) model without much further complication apart from a rapid increase in the number of parameters that need to be estimated.

VAR(p) analysis



Are we done with the stable VAR(p) ? Almost !!

For a given VAR(p) model, we can carry out a statistical dynamic analysis of multiple time series simultaneously.

The only question that remains (there are of course many more, but these are for another course !!):

How to determine p ?

We discuss two approaches:

- ▶ likelihood-ratio (LR) test;
- ▶ information criteria AIC.

Determine order in VAR: LR test

Suppose N time series are modeled by VAR(1) and VAR(2), both providing reasonable descriptions of the dynamics, which one to choose ?

We notice that VAR(1) is a special case of VAR(2) :

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} + \begin{bmatrix} \phi_{1,11} & \phi_{1,12} \\ \phi_{1,21} & \phi_{1,22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \\ \begin{bmatrix} \phi_{2,11} & \phi_{2,12} \\ \phi_{2,21} & \phi_{2,22} \end{bmatrix} \begin{pmatrix} y_{1,t-2} \\ y_{2,t-2} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

with $\phi_{2,11} = \phi_{2,12} = \phi_{2,21} = \phi_{2,22} = 0$.

Determine order in VAR: LR test

Hence we estimate parameters in a VAR(2) and test the null $H_0 : \phi_{2,11} = \phi_{2,12} = \phi_{2,21} = \phi_{2,22} = 0$ of a VAR(1) using the likelihood-ratio (LR) test:

$$2(\hat{\ell}_1^{\max} - \hat{\ell}_0^{\max}) = (T - 1) \left(\log |\hat{\Sigma}_{\text{MLE},1}| - \log |\hat{\Sigma}_{\text{MLE},0}| \right) \stackrel{a}{\sim} \chi_m^2,$$

with $m = N^2$, where $\hat{\ell}_1^{\max}$, $\hat{\Sigma}_{\text{MLE},1}$ are for VAR(2) and $\hat{\ell}_0^{\max}$, $\hat{\Sigma}_{\text{MLE},0}$ are for VAR(1) model.

This test can be used for any VAR(p) with choices of $p = p_1$ and $p = p_0$ with $p_1 > p_0$. Many combinations can be considered.

Determine order in VAR: AIC

The use of the Akaike information criterion (AIC) fits the VAR analysis very naturally. Its use is also similar to univariate autoregressive and related models.

The AIC selection procedure starts with a reasonably large $p = p_*$. Then all VAR(p_j) models are considered for each $p_j = 0, 1, \dots, p_*$ where VAR(0) is model $y_t = \omega + \varepsilon_t$:

- ▶ MLE of parameters leading to $\hat{\ell}_j^{\max}$, $\hat{\Sigma}_{\text{MLE},j}$ with index j corresponding to $p = p_j$;
- ▶ Compute AIC as given by

$$\text{AIC}_j = \log |\hat{\Sigma}_{\text{MLE},j}| + 2 p_j \cdot \frac{N^2}{T};$$

- ▶ Rank j according to AIC_j and choose p_j with smallest AIC.

Determine order in VAR: AIC

The use of the Akaike information criterion (AIC) fits the VAR analysis very naturally. Its use is also similar to univariate autoregressive and related models.

- ▶ AIC procedure is uniform and more clear when compared to the LR procedure.
- ▶ Make sure that p_* is sufficiently large: if AIC for p_* is smallest, then consider higher p_* .
- ▶ There are other information criteria: SBIC is an example.
- ▶ More advanced model selection procedures consider out-of-sample criteria and are based on estimation-validation sample splits.

What did we do?

We have investigated VAR models with a focus on

- ▶ dynamic properties (Wold decomposition, unconditional moments),
- ▶ forecasting (conditional moments),
- ▶ Granger causality,
- ▶ impulse response functions,
- ▶ parameter estimation by the (conditional) maximum likelihood method,
- ▶ model selection (order p , AIC).

General Theorem on Stationarity

Theorem: Let $\{Y_t\}$ be a weakly stationary (univariate) time series with $E(Y_t) = 0$ and autocovariance function γ_Y .

If $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, then the process

$$X_t = \sum_{j=-\infty}^{\infty} \psi_j Y_{t-j} = \psi(L)Y_t$$

is weakly stationary with mean zero and autocovariance function

$$\gamma_X(h) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \gamma_Y(h + k - j).$$

Proof of General Theorem

$$E(X_t) = E\left(\sum_{j=-\infty}^{\infty} \psi_j Y_{t-j}\right) = \sum_{j=-\infty}^{\infty} \psi_j E(Y_{t-j}) = 0.$$

$$\begin{aligned} \text{Cov}(X_{t+h}, X_t) &= E(X_{t+h} X_t) = E\left[\left(\sum_{j=-\infty}^{\infty} \psi_j Y_{t+h-j}\right)\left(\sum_{k=-\infty}^{\infty} \psi_k Y_{t-k}\right)\right] \\ &= \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k E(Y_{t+h-j} Y_{t-k}) \end{aligned}$$

Then by the weak stationarity of $\{Y_t\}$ we have

$$\text{Cov}(X_{t+h}, X_t) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} \psi_j \psi_k \gamma_Y(h-j+k),$$

which does not change with t !